

Gray's WMS & AI Digital Employee

Advanced & Unique Feature Roadmap

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The unfair advantage

Two assets sit right next to the warehouse screens that most WMS products do not have. The strongest features are the ones that combine them:

1. Guarded SQL gateway	The AI can actually read and write to the database through ai/executequery (reads) and ai/executewrite (single-statement, verb-whitelisted writes) — safely, with a confirm step.
2. Activity / task-mining log	Every navigation, click, screen dwell and now query parameter is recorded (WMS_ACTIVITY_LOG). The system can see what people really do, not just what the schema stores.

Tier 1 — Build next (highest leverage, reuses what exists)

These three turn the Digital Employee from a chat box into something that notices, acts, and learns.

1 “Watch me, then do it for me” — automation from the activity log

Mine a repeated multi-step sequence (already detected in Repetitive Works) and let the Digital Employee offer to automate it: “You’ve run Load Trips → select → Print 14 times this week — run this every morning at 8am?” One yes turns it into a scheduled job using the existing query/write path.

Why it's different: No other WMS turns observed behaviour into runnable automations. You're ~70% there already.

2 Exception & anomaly radar (proactive, not a dashboard)

A background pass over inventory / trips / receiving that surfaces only what's wrong: negative stock, trips with no printed docs, receipts open > N days, SKUs below reorder, printers failing jobs. Delivered as a morning briefing card + the header bell, each with an AI-written “why this matters + one-click fix.”

Why it's different: Warehouses live on exceptions; most systems make you go hunting for them.

3 Natural-language → action, with a confirm step

“Reprint all docs for trip 4471”, “which orders today have no PDF yet?”, “block bay B-12”. The AI drafts the SQL/action, shows the exact statement and a row preview, you approve, it runs through the guarded write.

Why it's different: The whole pipe already exists — this is mostly prompt work plus a confirm UI.

Tier 2 — Genuinely differentiating

Standalone wins that deepen the platform.

4 Digital Employee shift report

At end of day, read the activity log + data changes and write a plain-English summary: what got done, what's stuck, who was blocked, tomorrow's risks. Auto-emailed.

Why it's different: Turns raw logs into a management artefact nobody has to compile by hand.

5 Time-&-motion / friction coaching

Rage-clicks, long dwell and repeated searches are already detected. Turn them into a weekly nudge: "6h spent re-searching the same trips — here's a saved filter / a Forms Designer form that removes 4 clicks."

Why it's different: Feeds the activity log straight back into the Forms Designer to remove the friction it finds.

6 Voice-first floor mode

Push-to-talk (plumbing already built): "received 40 cases of SKU 8891 into A-3" → AI parses → confirm card → write. Hands-free receiving / put-away.

Why it's different: Huge on a floor where hands are full of gloves and scanners.

7 Smart print routing

Learn from history which printer / label each trip or doc type actually used, and auto-suggest it in the Print Job Queue.

Why it's different: Kills a recurring manual step using data you already log.

8 "Explain this record"

Right-click any trip / order / SKU → the AI assembles its full story from several queries (status, docs, history, related receipts) into one narrative.

Why it's different: Great for training and for settling disputes fast.

Tier 3 — Bigger platform bets

Longer builds; strategic.

9	Import from APEX → auto-generate WMS screens Use the metadata script to convert existing APEX pages into Forms Designer definitions — a migration tool.
10	Predictive workload Forecast tomorrow's trip / receiving volume and staffing from historical patterns.
11	Photo intake (vision) Snap a damaged pallet or delivery note; Claude vision extracts SKU / qty / condition into a receiving draft.
12	Digital-twin heat-map Colour a warehouse floor map by activity / pick density from the log (the route map, on the physical layout).

Recommended order & enablers

Build in this order: **#2 Exception Radar** → **#3 NL-to-action** → **#1 Automation-from-behaviour**.
Each reuses the gateways and the log, each ships standalone, and together they deliver the actual promise of the “Digital Employee” name.

Enabler	Why it gates the smart features
Real usernames	Activity currently logs as UNKNOWN. Per-person patterns, coaching and shift reports are worthless without it. Need the WMS logged-in-user variable / storage key.
A small scheduler	Background jobs (exception scan, shift report, automations) need a WinForms timer or an APEX scheduled job.

Gray's WMS — internal working document. Ideas captured for upcoming development; not committed scope.